Ilnur Baibulov

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Research Interests

- Mathematical physics
- Operator algebras

Education

2014 – 2018 B.Sc. in Physics, Saint Petersburg State University, Faculty of Physics

2018 – 2020 M.Sc. in Physics, Saint Petersburg State University, Faculty of Physics

2021 – 2025 PhD studies, Saint Petersburg State University, Faculty of Physics

Skills

Languages

Russian (native), English (fluent), French (reading with a dictionary)

Software

LaTeX, Matlab, Wolfram Mathematica, Sage, Python, Lean

PhD Thesis

2026 The localization principle on quasi-orbits in C^*-crossed products and its application to the C^*-algebra of one-dimensional singular integral operators with semi-almost periodic coefficients.

Supervisor: *O. V. Sarafanov*

Publications and Talks

Journal Articles

2026

- **I. V. Baibulov**, The primitive spectrum of the C^*-algebra of singular integral operators on a complex contour with discontinuous semi-almost periodic coefficients, *Algebra i Analiz*, 38:1 (2026), 40–65 (in Russian).
  https://www.mathnet.ru/eng/aa1988.
- **I. V. Baibulov**, On the principle of localization on quasi-orbits in crossed products, *Mathematical Notes*, 119:4 (2026), 559–574.
  https://doi.org/10.1134/S0001434626602030.
  Russian original: *Matematicheskie Zametki*, 119:4 (2026), 471–491, https://doi.org/10.4213/mzm14787.

2025

- **I. V. Baibulov**, On the representations of the C^*-algebra of singular integral operators on a complex contour with discontinuous semi-almost periodic coefficients, *Izvestiya: Mathematics*, 89:6 (2025), 1125–1162.
  https://doi.org/10.4213/im9665e.
  Russian original: *Izvestiya RAN. Seriya Matematicheskaya*, 89:6 (2025), 45–84,
  https://doi.org/10.4213/im9665.

2024

- **I. V. Baibulov**, O. V. Sarafanov, The spectrum of the C^*-algebra of singular integral operators with semi-almost periodic coefficients, *Sbornik: Mathematics*, 215:4 (2024), 464–493.
  https://doi.org/10.4213/sm9999e.
  Russian original: *Matematicheskii Sbornik*, 215:4 (2024), 30–61, https://doi.org/10.4213/sm9999.

2023

- **I. V. Baibulov**, On Justification of the Asymptotics of Three One-Dimensional Short-Range Particles Scattering Problem Solution for the Processes $3 \to 2$, *Journal of Mathematical Sciences*, 277 (2023), 500–505.
  https://doi.org/10.1007/s10958-023-06857-5.
  Russian original: *Zapiski Nauchnykh Seminarov POMI*, 493 (2020), 40–47.

2021

- **I. V. Baibulov**, A. M. Budylin, S. B. Levin, The Scattering Problem of Three One-Dimensional Short-Range Quantum Particles Involving Bound States in Pair Subsystems. The Coordinate Asymptotics of the Resolvent Kernel and of the Eigenfunctions of the Absolutely Continuous Spectrum, *Journal of Mathematical Sciences*, 252 (2021), 567–575.
  https://doi.org/10.1007/s10958-021-05181-0.
  Russian original: *Zapiski Nauchnykh Seminarov POMI*, 483 (2019), 5–18.

2019

- **I. V. Baibulov**, A. M. Budylin, S. B. Levin, The Asymptotics of Eigenfunctions of the Absolutely Continuous Spectrum. The Scattering Problem of Three One-Dimensional Quantum Particles, *Journal of Mathematical Sciences*, 243 (2019), 640–655.
  https://doi.org/10.1007/s10958-019-04566-6.
  Russian original: *Zapiski Nauchnykh Seminarov POMI*, 471 (2018), 15–37.
- **I. V. Baibulov**, A. M. Budylin, S. B. Levin, On Justification of the Asymptotics of Eigenfunctions of the Absolutely Continuous Spectrum in the Problem of Three One-Dimensional Short-Range Quantum Particles with Repulsion, *Journal of Mathematical Sciences*, 238 (2019), 566–590.
  https://doi.org/10.1007/s10958-019-04258-1.
  Russian original: *Zapiski Nauchnykh Seminarov POMI*, 461 (2017), 14–51.

Selected Talks

2025 Seminar on Operator Theory and Function Theory (St. Petersburg Department of Steklov Mathematical Institute of RAS, Saint Petersburg).

V. M. Babich Seminar on Diffraction Theory and Wave Propagation (St. Petersburg Department of Steklov Mathematical Institute of RAS, Saint Petersburg).

2024 Noncommutative Geometry in NYC (St John's University, New York City), *online talk*.

Seminar of the Department of Higher Mathematics and Mathematical Physics, Faculty of Physics, SPbU (Saint Petersburg).

2023 Seminar on Noncommutative Geometry and Topology (Lomonosov Moscow State University, Faculty of Mechanics and Mathematics, Moscow), *online talk*.

Third Conference of Russian Mathematical Centers (October 10–15, 2023, Maykop).

Saint Petersburg Winter Youth Conference on Probability Theory and Mathematical Physics (December 4–7, 2023, Saint Petersburg).

2020 Seminar in Spectral Theory and Related Topics, Leonhard Euler International Mathematical Institute, *online talk*.

2019 V. M. Babich Seminar on Diffraction Theory and Wave Propagation (St. Petersburg Department of Steklov Mathematical Institute of RAS, Saint Petersburg).

2018 Seminar of the Laboratory of Few-Body Systems (Bogoliubov Laboratory of Theoretical Physics, JINR, Dubna).

Grants and Scholarships

2025 – 2026 Russian Science Foundation grant No. 22-11-00070-P (research group member).

2021 – 2024 Leonhard Euler International Mathematical Institute / SPbU, Saint Petersburg, support programme for postgraduate students of Saint Petersburg.

2023 – 2024 Russian Science Foundation grant No. 22-11-00070 (research group member).

2020 Leonhard Euler International Mathematical Institute / SPbU, Saint Petersburg, support programme for students of Saint Petersburg.

2018 – 2019 O. A. Ladyzhenskaya Scholarship of the St. Petersburg Mathematical Society.

2018 – 2020 Russian Science Foundation grant No. 17-11-01003 (research group member).

Employment History

05 / 2026 – present Chebyshev Laboratory, SPbU, Saint Petersburg – *research engineer.*

09 / 2024 – 05 / 2025 SPbU, Faculty of Physics, Saint Petersburg – *problem sessions and elective lectures.*

12 / 2021 – 12 / 2025 Leonhard Euler International Mathematical Institute, SPbU, Saint Petersburg – *research engineer.*

08 / 2022 Shenzhen MSU–BIT University, Shenzhen – *online course on ordinary differential equations.*

09 / 2021 – 05 / 2022 SPbU, Graduate School of Management, Saint Petersburg – *lectures and problem sessions in mathematics.*

04 / 2021 School "Spectral and Functional Inequalities and Their Applications", Sirius Mathematics Center, Sochi – *teaching assistant.*

07 / 2020 – 12 / 2020 Leonhard Euler International Mathematical Institute / SPbU, Saint Petersburg – *research engineer.*

12 / 2019 The Royal Swedish Academy of Sciences, Institut Mittag-Leffler, Djursholm, Sweden – *research visit.*

04 / 2018 – 12 / 2019 SPbU, Faculty of Mathematics and Mechanics, Saint Petersburg – *research assistant, grant participant.*

09 / 2016 – 05 / 2017 Presidential Physics and Mathematics Lyceum No. 239, Saint Petersburg – *teacher of extracurricular classes.*